

A DERIVATION OF THE THICK SPRAY MODEL

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My website

I recently completed my PhD and am actively seeking postdoctoral opportunities.



Motivation and spray regimes

Sprays are two-phase flows in which dispersed particles/droplets interact with a surrounding gas.

We classify sprays according to the dispersed phase volume fraction.

- **Very thin sprays:** volume fraction $\ll 10^{-3}$.
- **Thin sprays:** volume fraction $\ll 10^{-1}$; drag effects scale like $\mathcal{O}(N_p r_p^2)$ (proportional to surface area).
- **Thick sprays:** volume fraction up to about 10^{-1} ; finite-volume effects produce pressure corrections of order $\mathcal{O}(N_p r_p^3)$.

Many models can be used to describe such systems. In this work, we formally derive a kinetic–fluid description (Vlasov–Euler) from a kinetic–kinetic description based on Enskog–Boltzmann operators.

Enskog equation and finite-size correction

The Boltzmann equation describes dilute gases, in which particle volume is neglected. In the present work, finite-size effects are retained through the Enskog correction. For a single-species gas, the Enskog operator takes the form

$$\partial_t F(t, x, v) + v \cdot \nabla_x F(t, x, v) = \mathcal{E}[F, F],$$

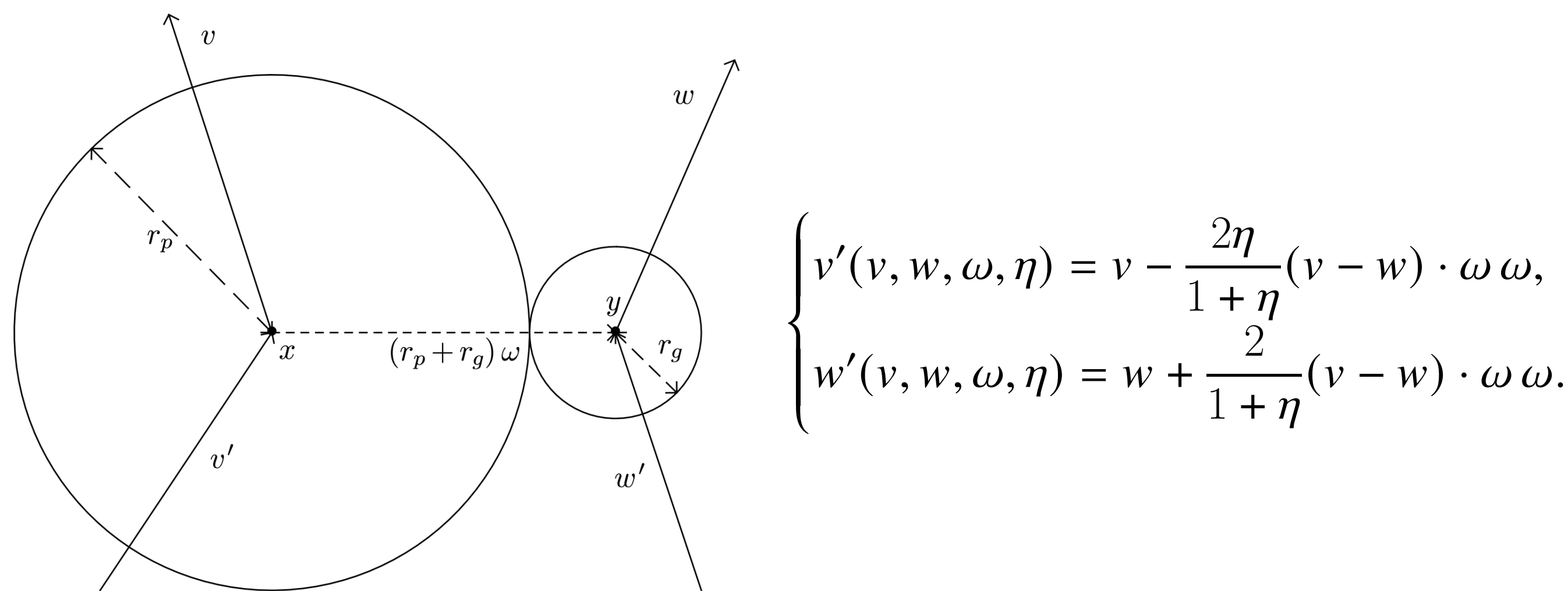
with

$$\mathcal{E}[g, h](t, x, v) = d^2 \int_{(v-v_*) \cdot \omega > 0} \left[h(t, x + d\omega, v'_*(v, v_*, \omega, 1)) g(t, x, v'(v, v_*, \omega, 1)) - h(t, x - d\omega, v_*) g(t, x, v) \right] (v - v_*) \cdot \omega d\omega dv_*,$$

where d denotes the diameter of particles.

Cross-species collision geometry

In the binary gas–particle setting, we use the cross-species Enskog–Boltzmann operators $\mathcal{E}_1^\eta[f, F]$ and $\mathcal{E}_2^\eta[F, f]$, with shifts $x \pm (r_g + r_p)\omega$.



Compact parameter table

Parameter	Meaning
L	container size (periodic box)
T	time scale
N_p, N_g	number of particles and gas molecules $/L^3$
V_p, V_g	thermal speeds of particles and gas molecules
S_p, S_g	average collision cross sections
r_p, r_g	particle and gas radii
m_p, m_g	particle and gas masses
$\eta = m_g/m_p \leq 1$	mass ratio
$\epsilon = V_p/V_g \leq 1$	thermal speed ratio
$\delta = 1/(N_g L S_g)$	Knudsen number of gas phase

Kinetic starting point

After nondimensionalisation, and under the scaling assumptions

$$\epsilon = 1, \quad T = L/V_g, \quad L^3 N_p = 1, \quad N_p L S_p \ll 1, \quad r_g/L \ll 1,$$

the system becomes

$$\begin{cases} \partial_t f^{\eta, \delta} + w \cdot \nabla_x f^{\eta, \delta} = \mathcal{E}_1^\eta[f^{\eta, \delta}, F^{\eta, \delta}] + \frac{1}{\delta} \mathcal{B}[f^{\eta, \delta}, f^{\eta, \delta}], \\ \partial_t F^{\eta, \delta} + v \cdot \nabla_x F^{\eta, \delta} = \frac{1}{\eta} \mathcal{E}_2^\eta[F^{\eta, \delta}, f^{\eta, \delta}]. \end{cases} \quad (1)$$

Here $\mathcal{E}_1^\eta$ and $\mathcal{E}_2^\eta$ are Enskog–Boltzmann collision operators, while $\mathcal{B}$ is the classical Boltzmann collision operator.

Main formal result

Under the scaling and convergence assumptions, as $\delta, \eta \rightarrow 0$, suppose that

$$\begin{aligned} f^{\eta, \delta}(t, x, w) &\rightarrow \mu[\alpha(t, x) n(t, x), u(t, x), \theta(t, x)/m_g](w), \\ F^{\eta, \delta}(t, x, v) &\rightarrow F(t, x, v), \end{aligned}$$

where μ denotes the local Maxwellian $\mu[R, U, T](w) = \frac{R}{(2\pi T)^{3/2}} \exp\left(-\frac{|w-U|^2}{2T}\right)$. Then the limit functions $(\alpha, n, u, \theta, F)$ satisfy the following Euler–Vlasov equation for thick sprays. (For brevity, we omit the corresponding energy equation from the poster)

$$\begin{cases} \partial_t(\alpha p) + \operatorname{div}_x(\alpha p u) = 0, \\ \partial_t(\alpha p u) + \operatorname{div}_x(\alpha p u \otimes u) + \alpha \nabla_x p = m_g \int_{\mathbb{R}^3} F(v) D(v - u) dv \\ \quad - \operatorname{div}_x \left(\alpha n \int_{\mathbb{R}^3} F(v) Q(v - u) dv \right) + o(a^3), \\ \partial_t F + v \cdot \nabla_x F + \operatorname{div}_v \left(F \left[D(v - u) + \frac{4\pi}{3m_g} a^3 \nabla_x p \right] \right) = o(a^3), \\ \alpha(t, x) = 1 - \frac{4\pi}{3} a^3 \int_{\mathbb{R}^3} F(t, x, v) dv, \\ p(t, x) = n(t, x) \theta(t, x), \quad \rho(t, x) = m_g n(t, x). \end{cases} \quad (2)$$

The key novelty is not merely the recovery of the standard engineering model: the volume fraction α appears explicitly in front of the pressure gradient. The derivation also reveals higher-order force corrections of order $\mathcal{O}(a^3)$.

Idea of the derivation

The key idea is to remove the spatial delocalisation in the Enskog–Boltzmann operators by means of a first-order Taylor expansion:

$$f^{\eta, \delta}(x \pm a\omega, w) = f^{\eta, \delta}(x, w) \pm a\omega \cdot \nabla_x f^{\eta, \delta}(x, w) + o(a).$$

With this expansion, the collision operator $\mathcal{E}_2^\eta$ decomposes into a leading friction contribution and a higher-order correction:

$$\mathcal{E}_2^\eta[F^{\eta, \delta}, f^{\eta, \delta}] = \underbrace{\mathcal{R}_1^\eta[F^{\eta, \delta}, f^{\eta, \delta}]}_{\text{friction term}} - \underbrace{\mathcal{R}_2^\eta[F^{\eta, \delta}, f^{\eta, \delta}]}_{\text{buoyancy-type correction}} + o(a^3).$$

The coefficient $\frac{4\pi}{3}a^3$ in the definition of α ultimately comes from two isotropy mechanisms: the angular integration in ω , and the fact that the Maxwellian depends only on $|w - u|$. In particular,

$$2a^3 \int_{|\omega|=1, \omega \cdot (v-w) > 0} (\omega \cdot (v - w))^2 \omega^{\otimes 2} d\omega = \frac{4\pi}{15} a^3 (2(v - w)^{\otimes 2} + |v - w|^2 \operatorname{Id}).$$

This identity is precisely what produces the finite-volume correction in the macroscopic limit.

Essential references

1. Z. Chen. A Formal Derivation of the Thick Spray Model from the Enskog–Boltzmann System. *arXiv:2509.05118*, 2025.
2. L. Desvillettes, F. Golse, and V. Ricci. A formal passage from a system of Boltzmann equations for mixtures towards a Vlasov–Euler system of compressible fluids. *Acta Mathematicae Applicatae Sinica, English Series*, 35(1):158–173, 2019.
3. E. Bernard, L. Desvillettes, F. Golse, and V. Ricci. A derivation of the Vlasov–Stokes system for aerosol flows from the kinetic theory of binary gas mixtures. *Kinetic and Related Models*, 11(2):227–250, 2018.
4. L. Desvillettes, E. Bernard, F. Golse, and V. Ricci. A derivation of the Vlasov–Navier–Stokes model for aerosol flows from kinetic theory. *Communications in Mathematical Sciences*, 15(6):1703–1741, 2017.
5. P. J. O’Rourke. Collective drop effects on vaporizing liquid sprays. *PhD Thesis, Los Alamos National Laboratory*, 1981.